



## 0. Introduction

### 1. Pre-requisites for Full Closed Loop

- 1.1 Well tuned hybrid closed loop
- 1.2 Fast insulin
- 1.3 Reliable insulin delivery from pump and cannula
- 1.4 Excellent CGM
- 1.5 Meal-related limitations?
- 1.6 Lifestyle-related limitations?
- 1.7 Time required for setting-up

Case study 1.1: Occlusion

Case study 1.2: Comparing insulins for FCL

Case study 1.3: Jumpy CGM

Case study 1.4: Lost pump connection

Case study 1.5: Permanent CGM values w/ 2x G6

Case study 1.6: When catastrophe strikes

Case study 1.7: MDI when the loop dies...

Case study 1.8: (placeholder for a Libre3 / 1.-minute case)...

...

### 2. General Settings for Full Closed Loop

- 2.1 SMB range extension
- 2.2 Max and min autoISF ratio
- 2.3 SMB delivery ratio
- 2.4 iobTH (iob\_threshold\_percent)
- 2.5 Eating Soon TT?
- 2.6 General settings in AAPS/Preferences

### 3. Description of autoISF / guidance by developers

- 3.1 Overview
- 3.2 ISF modulation flowcharts
- 3.3 Exercise mode and dynamic iobTH
- 3.4 Automation options with autoISF parameters
- 3.5 Activity monitor
- 3.6 Using one-minute CGM (Libre 3)
- 3.7 AutoISF parameters overview table
- 3.8 Emulator for logfile analysis and tuning
- 3.9 Links to related case studies/detailed doc.s

### 4. Meals: Setting ISF\_weights in AAPS/Preferences

- 4.1 Getting started
- 4.2 bgAccel\_ISF\_weight
- 4.3 pp\_ISF\_weight
- 4.4 bgBrake\_ISF\_weight
- 4.5 dura\_ISF and bg\_ISF
- 4.6 Tuning your initial settings
- 4.7 Complex scenarios
- 4.8 Profile helper

Case study 4.1: Pizza

Case study 4.2: Low carb meals

Case study 4.3: Hands-off FCL around Christmas

### 5. Temp. modulation of autoISF aggressiveness

#### 5.1 Automatic modulation of loop aggressiveness

- 5.1.1 autoISF off outside of meal windows

|     |                                                                              |
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| 59  | 5.1.3 SMB off @ odd temp. target                                             |
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